

COSC/MATH 4340: Numerical Methods for Differential Equations

Numerical Linear Algebra (NLA): Spectral Methods

Class: COSC/MATH 4340

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Spectral Decompositions

Definition

Let A be a square, symmetric matrix of size n . λ_i and $\mathbf{v}_i$ are an *eigenvalue* and *eigenvector* of A if the following is true

$$A\mathbf{v}_i = \lambda_i\mathbf{v}_i$$

Definition

Each λ_i is a root of the *characteristic polynomial*

$$p(\lambda) = \det(A - \lambda I)$$

Note: Each $\mathbf{v}_i$ is in the *null space* of matrix $(A - \lambda_i I)$ and is unique up to normalization and a sign change.

(See notes for example.)



Eigenvalues and Eigenvectors

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Spectral Decompositions

Definition

Matrix A is *diagonalizable* if it has n eigenvalues (possibly repeated) and n linearly independent eigenvectors $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_N\}$. Define the matrices $D = \text{diag}[\lambda_1, \dots, \lambda_N]$ and $V = [\mathbf{v}_1, \dots, \mathbf{v}_N]$, then the *spectral decomposition* of A is given by

$$A = VDV^{-1} = \sum_j \lambda_j \mathbf{v}_j \mathbf{u}_j^T,$$

where $V^{-1} = U^T$ and $U = [\mathbf{u}_1, \dots, \mathbf{u}_N]$.

(See notes and code.)



Matrix Exponentials

Note: For any positive integer $k \geq 0$ one has

$$A^k = VD^kV^{-1} = \sum_j \lambda_j^k \mathbf{v}_j \mathbf{u}_j^T,$$

Definition

If matrix A is diagonalizable, then its *matrix exponential* is given by

$$\begin{aligned} e^A &= I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots \\ &= V(I + D + \frac{D^2}{2!} + \frac{D^3}{3!} + \dots)V^{-1} \\ &= Ve^D V^{-1} \\ &= \sum_j e^{\lambda_j} \mathbf{v}_j \mathbf{u}_j^T. \end{aligned}$$



Spectral Solution to Linear Systems of ODEs

Theorem

Consider the linear system of ODEs

$$\frac{d}{dt}\mathbf{x} = A\mathbf{x}$$

with initial condition $\mathbf{x}(0)$. If A is a square diagonalizable matrix, then the solution $\mathbf{x}(t)$ is given by

$$\mathbf{x}(t) = e^{tA}\mathbf{x}(0) = Ve^{tD}V^{-1}\mathbf{x}(0) = \sum_j c_j e^{t\lambda_j} \mathbf{v}_j,$$

where $c_j = [V^{-1}\mathbf{x}(0)]_j = [U^T\mathbf{x}(0)]_j = \mathbf{u}_j^T\mathbf{x}(0)$.

(See notes and code.)



Equilibrium and Stability of Systems of ODEs

Definition

If $\frac{d}{dt}\mathbf{x} = F(\mathbf{x})$ is a system of ODEs, then vector $\mathbf{x}^*$ is an *equilibrium* iff $F(\mathbf{x}^*) = \mathbf{0}$.

Lemma

For a linear system of ODEs, $F(\mathbf{x}) = A\mathbf{x}$, and $A\mathbf{x} = \mathbf{0}$ iff $\mathbf{x} \in \text{Null}(A)$. If A is diagonalizable, then $\text{Null}(A) = \mathbf{0}$ and so $\mathbf{x}^ = \mathbf{0}$ is the only equilibrium.*

Equilibrium and Stability of Systems of ODEs

Definition

An equilibrium of an ODE is called *asymptotically stable* if there exists a region $\mathcal{S} \in \mathbb{R}^n$ containing $\mathbf{x}^*$ such that $\mathbf{x}(t) \rightarrow \mathbf{x}^*$ for any initial condition $\mathbf{x}(0)$ within $\mathcal{S}$.

Definition

An equilibrium is *globally* asymptotically stable if $\mathbf{x}(t) \rightarrow \mathbf{x}^*$ for any $\mathbf{x}(0) \in \mathbb{R}^n$. It's *locally* asymptotically stable if $\mathbf{x}(t) \rightarrow \mathbf{x}^*$ only for a strict subset $\mathcal{S} \subset \mathbb{R}^n$.

Definition

An equilibrium is *unstable* if there exists initial conditions arbitrarily close to $\mathbf{x}^*$ that diverge from $\mathbf{x}^*$.



Equilibrium and Stability of Systems of ODEs

Lemma

The equilibrium $\mathbf{x}^ = \mathbf{0}$ of a linear system of ODEs, $\frac{d}{dt}\mathbf{x} = A\mathbf{x}$, is asymptotically stable if and only if all eigenvalues λ_i of A are negative.*

Lemma

The equilibrium $\mathbf{x}^ = \mathbf{0}$ is unstable stable if and only if one (or more) eigenvalue is positive.*

Lemma

If A has both positive and negative eigenvalues, then $\mathbf{x}^ = \mathbf{0}$ is called a saddle point.*



Robustness of Eigenvalues and Eigenvectors

Given the importance of matrix spectral properties for solving ODEs, it's important to

- ▶ understand their robustness to perturbations/error
- ▶ develop efficient methods to approximately find them.

Theorem (Bauer-Fike)

Let A be a diagonalizable matrix, $A = VDV^{-1}$. Let E be a perturbation to A , and let λ be an eigenvalue of matrix $A + E$. Then

$$\min_i |\lambda - \lambda_i| \leq \text{cond}(V) \|E\|.$$

(See notes for proof.)

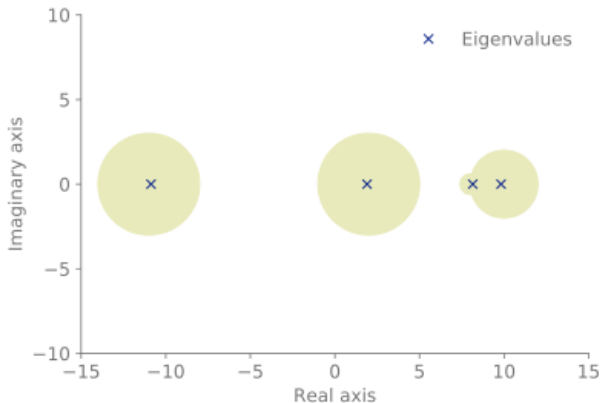


Robustness of Eigenvalues and Eigenvectors

Theorem (Gerschgorin Circles)

Let A be square matrix with diagonal entries $\{A_{ii}\}$ and for each i define a circle $\mathcal{C}_i = \{z; |z - A_{ii}| < r_i\}$ in the complex plane $\mathbb{C}$, where radius $r_i = \sum_{j \neq i} |A_{ij}|$ is determined by the size of the off-diagonal entries. Then each eigenvalue λ_i of A belongs to one of the circles, and if m circles are overlapping then exactly m eigenvalues are found in their union.

Robustness of Eigenvalues and Eigenvectors



(See notes for example.)



Robustness of Eigenvalues and Eigenvectors

Theorem (Matrix Spectral Perturbations)

Let $A(\epsilon) = A + \epsilon B$ be a diagonalizable matrix, $A = VDV^{-1}$, where $D = \text{diag}[\lambda_1, \dots, \lambda_N]$, $V = [\mathbf{v}_1, \dots, \mathbf{v}_N]$ and $V^{-1} = U^T$ with $U = [\mathbf{u}_1, \dots, \mathbf{u}_N]$. Assume the eigenvalues are non-repeated. Then one can approximate the eigenvalues and eigenvectors of $A(\epsilon)$ by

$$\lambda_k(\epsilon) = \lambda_k + \epsilon \frac{\mathbf{u}_k^T B \mathbf{v}_k}{\mathbf{u}_k^T \mathbf{v}_k} + \mathcal{O}(\epsilon^2)$$

$$\mathbf{v}_k(\epsilon) = \mathbf{v}_k + \epsilon a_k \mathbf{u}_k + \epsilon \sum_{j \neq k} \frac{\mathbf{u}_j^T B \mathbf{v}_k}{(\lambda_k - \lambda_j) \mathbf{u}_j^T \mathbf{v}_j} \mathbf{u}_j + \mathcal{O}(\epsilon^2)$$

(See notes for sketch of proof.)